



NED University of Engineering and Technology
Department of Computer and Information Systems Engineering
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BE (Computer Systems) Program

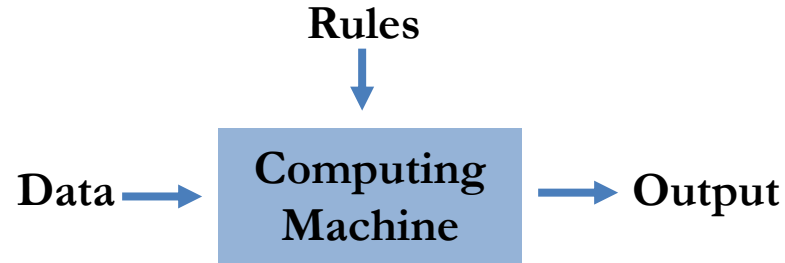
CS-324 Machine Learning

Introduction to Machine Learning

Dr. Maria Waqas

Traditional Approach

- You tell a computer what to do



Example

- Identifying Apples**

Setting rules to identify apples:

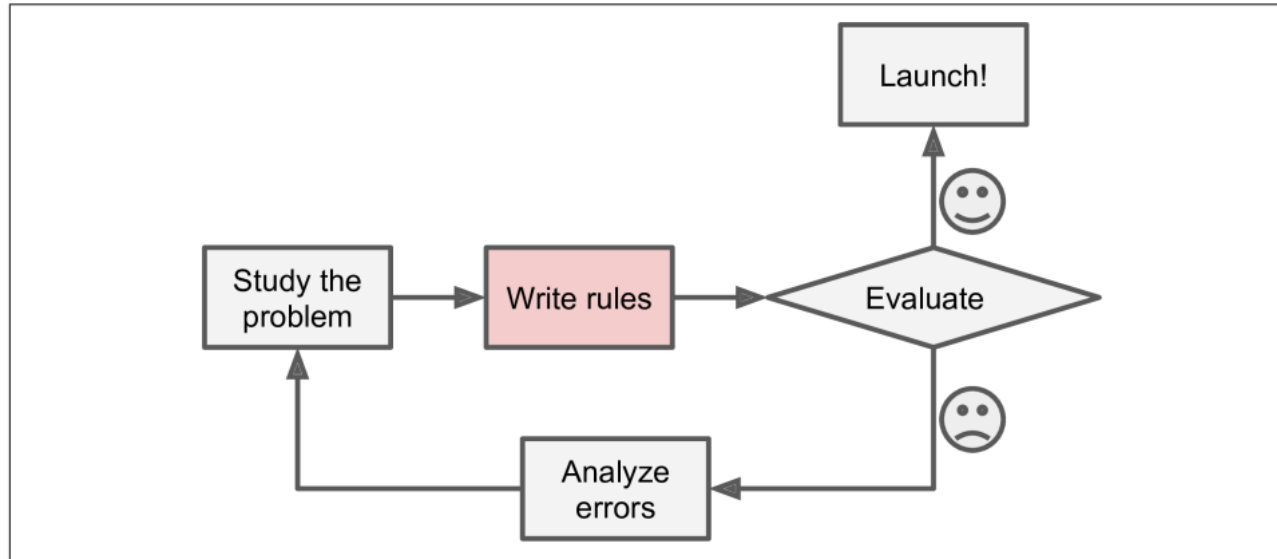
```
if color=red:  
    'Apple it is!'
```

```
if shape=round and color=red and edible  
    'Apple it is!'
```



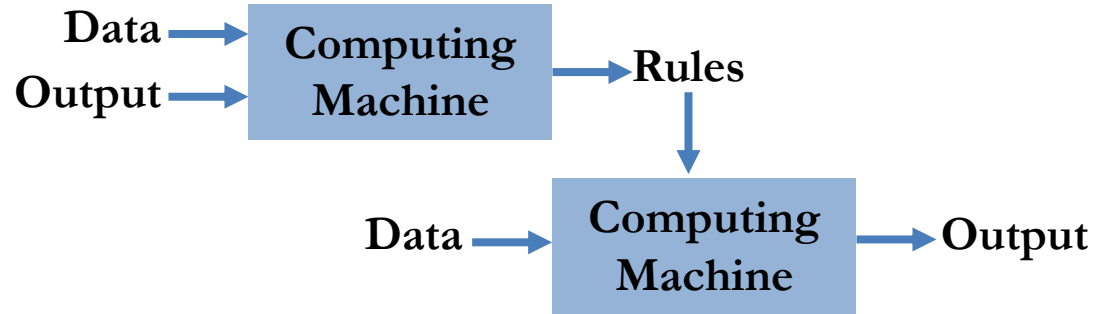


Traditional Approach



Machine Learning Approach

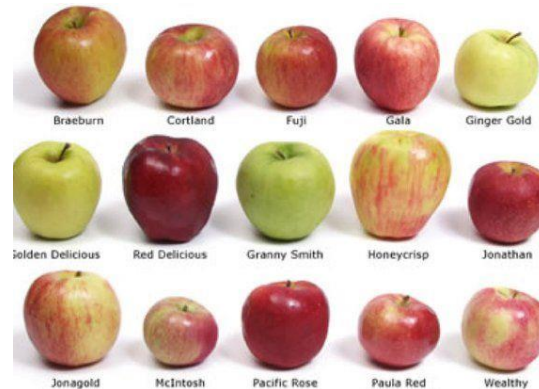
- You show a computer how to do a task
- Mimics how kids learn: we show them apples of different colors, of different shapes, from different angles, etc.



Example

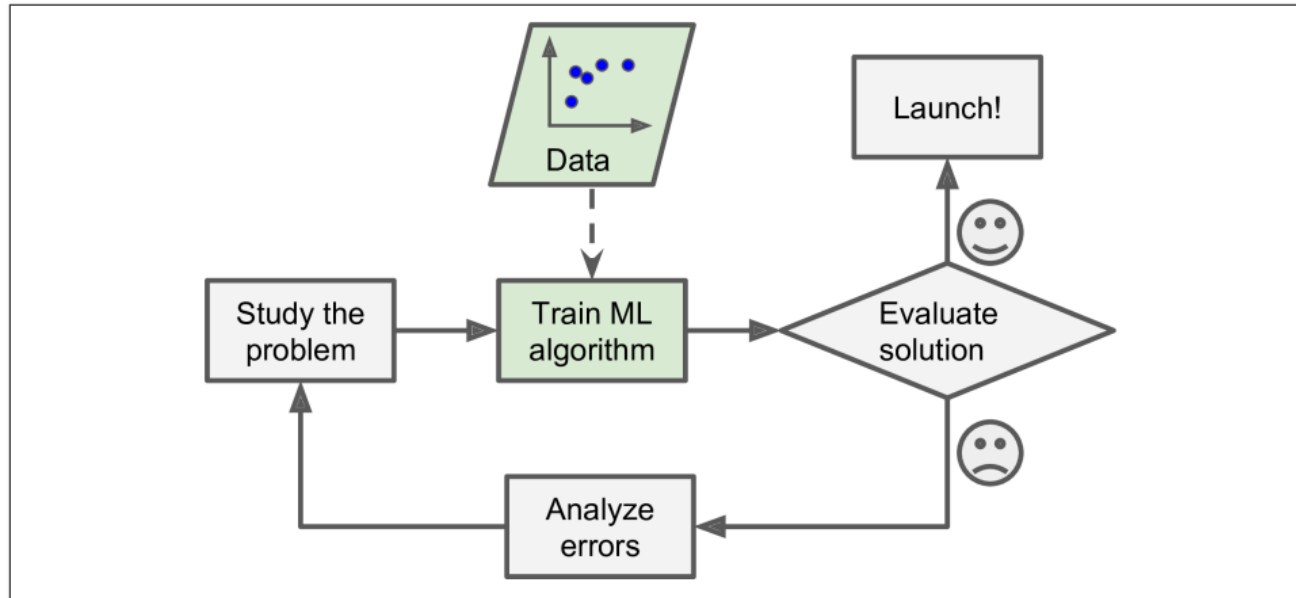
■ Identifying Apples

Showing multiple pictures of apples to the machine.





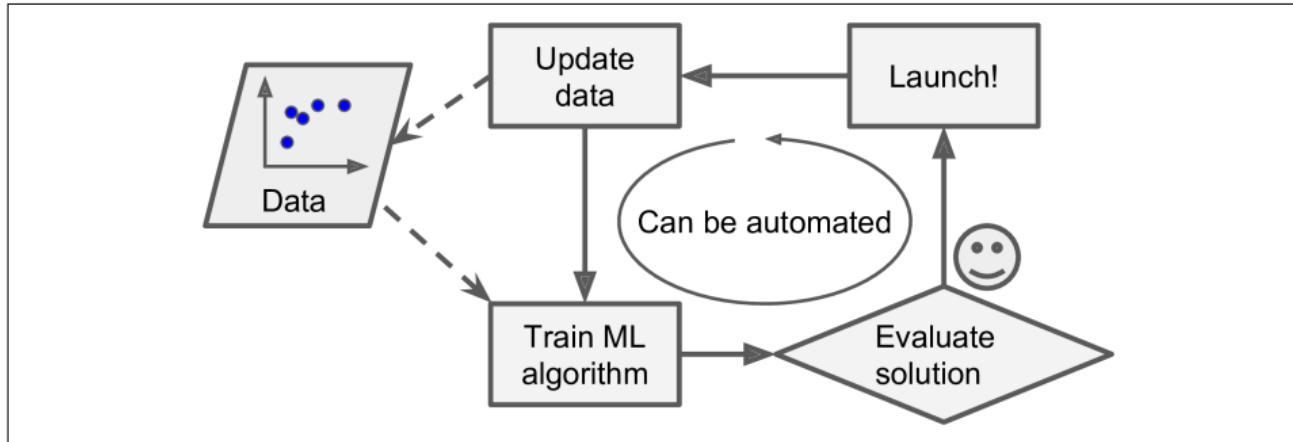
Machine Learning Approach





Why Use Machine Learning

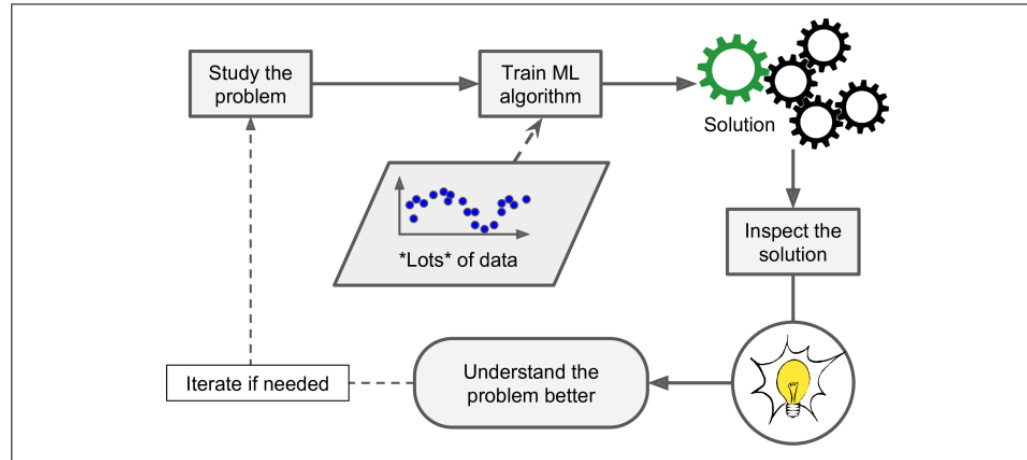
1. Problems for which existing solutions require a lot of hand-tuning or long lists of rules: one machine learning algorithm can often simplify code and perform better.
2. Complex problems for which there is no good solution at all using a traditional approach.
 - Tasks that fall into this category include multiple objects detection and localization, automatic speech recognition and other non-linear problems.
3. Fluctuating environments.
 - A machine learning algorithm can be made to automatically adapt to change.





Why Use Machine Learning

4. Getting insights about complex problems and large amounts of data.
- Finally, Machine Learning can help humans learn.
 - Applying ML techniques to dig into large amounts of data can help discover patterns that were not immediately apparent. This is called data mining.





Machine Learning Everywhere!

- Writing computer programs to do interesting stuff involving learning is a tough thing to do.
- It was realized that the only way to do this is by making the machine learn by itself.

- A computer observes some data, builds a model based on the data, and uses the model as a piece of software that can solve problems.

- A diverse range of learning models are available, the key is to choose the right one for a problem at hand.
- Many scientists think the best way to make progress on this is through learning algorithms called neural networks, which mimic how the human brain works.

- Applications based on machine learning are ubiquitous now a days.
 - Search engines like Google or Bing, know how to rank web pages.
 - Facebook or Apple's photo application recognizes your friends in your pictures.
 - Email spam filters sort tons of spam and non-spam emails.
 - Medical records are analyzed to understand disease better.
 - Computational biology helps understand human genome better.



What is Machine Learning

Definition 1

- Machine learning is the science of getting computers to learn without being explicitly programmed.
(Arthur Samuel, 1959)
 - Samuel's checkers program

Definition 2:

- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .
(Tom M. Mitchell, 1997)



What is Machine Learning

Example 1

Playing checkers

- The task of playing checkers - **task T**
- The experience of playing many games of checkers - **experience E**
- The probability that the program will win the next game - **performance measure P**

Example 2

Classifying emails as spam or non-spam

Your email program watches which emails you do or do not mark as spam. So in an email client like this, you might click the Spam button to report some email as spam but not other emails; and based on which emails you mark as spam, say your email program learns better how to filter spam email. Identify T, P and E in this setting.

- Classifying emails (flag email as spam or non-spam) - **task T**
- Watching you label emails as spam or not spam (training set) - **experience E**
- The fraction of emails correctly classified - **performance measure P**



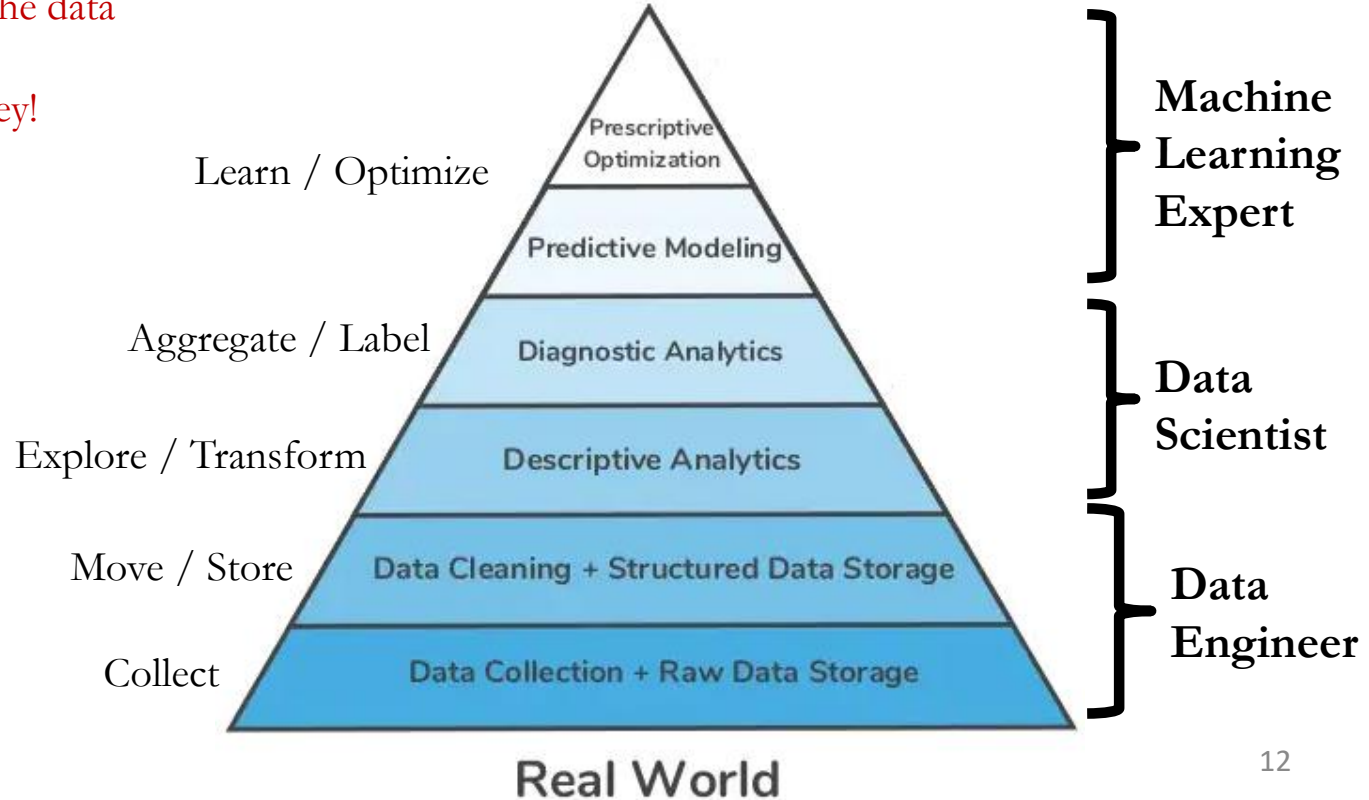
Data Science

- Tons of data are available
 - Clickstream data!
 - tons of data are available
- Turn data into knowledge to make use of it
 - Turn data into value!
- In 1980s IBM came out with the first relational databases storing things like customer details, payroll data.
- Soon after we started thinking about what useful things we can mine from data and the term datamining was coined.
- Datamining is the application of specific algorithms for extracting patterns from data.
- Extraction of patterns from data has traditionally been done with plain old statistics.
- Data mining + Computer Science = Data Science
- Make most of the data collected on every click, get useful insights using artificial intelligence and machine learning.



Data Science – Hierarchy of Needs

- Hierarchy of needs – Start simple and grow complexity over time.
- ML is only as good as the data you provide it.
- Quality of data is the key!





Addressing a Machine Learning Problem

Question Formulation

- Good design depends on good hypothesis or question.

Data Collection

- Collecting and preparing the dataset is very crucial while working on a machine learning project.
- Most machine learning algorithms cannot work properly without a well prepared and pre-processed dataset.
- Huge volumes of data are required for training of a machine learning algorithm.

Data Preprocessing

- Data Cleaning: Check for missing, incomplete and inaccurate data, format data.

Data Exploration and Visualization

- Ensure that the data represents what you think it represents.
- Spend time in inspecting and analyzing data.

Feature Engineering

- Feature extraction, labeling and aggregation.

Algorithm Training

- Linear regression, k-nearest neighbors, Bayes classifier, decision trees, etc.

Evaluation

- Choose the best model.



Types of Machine Learning Systems

- A wide variety of machine learning algorithms are available nowadays.
- Two common classification criteria are:
 1. Whether or not they are trained with human supervision (supervised, unsupervised, semi supervised, and reinforcement learning)
 2. Whether they are instance-based or model-based learning



Supervision in Learning

- Machine Learning systems can be classified according to the amount and type of supervision they get during training.
- There are four major categories:
 1. Supervised Learning
 2. Unsupervised Learning
 3. Semi-supervised Learning
 4. Reinforcement Learning



Supervised Learning

- In supervised learning, the training data fed to the algorithm includes the desired solutions, called labels.
- Given a data set, we already know what our correct output should look like, having the idea that there is a relationship between the input and the output.
- The agent observes input-output pairs and learns a function that maps from input to output.
- Supervised learning problems are categorized into regression and classification problems.

Regression

- In a regression problem, we try to predict results within a continuous output.
- Map input variables to some continuous function.
- The output can take any value.
- **Example:** Predicting the price of a car given a set of features like mileage, age, brand, etc.

Classification

- In a classification problem, we try to predict results in a discrete output.
- Map input variables into discrete categories.
- The output can take only some discrete values.
- **Example:** Classifying emails as spam or non-spam.



Unsupervised Learning

- In unsupervised learning the dataset is a collection of unlabeled examples.
- Unsupervised learning allows us to approach problems with little or no idea what our results should look like.
- The agent learns patterns in the input without any explicit feedback.
- Unsupervised learning algorithms can be categorized into clustering, anomaly detection and novelty detection, and association rule learning.

Clustering

- Clustering finds groups, having similar characteristics in unlabeled dataset.
- More generic in nature.
- **Example:** You may want to run a clustering algorithm on your blog visitors to try to detect groups of similar visitors. The algorithms may notice that 40% of your visitors are males who love comic books and generally read your blog in the evening, while 20% are young sci-fi lovers who visit during the weekends, and so on. If you use a hierarchical clustering algorithm, it may also subdivide each group into smaller groups. This may help you target your posts for each group.



Unsupervised Learning

Anomaly Detection and Novelty Detection

- The system is shown mostly normal instances during training, so the machine learning algorithm learns to recognize them and when it sees a new instance it can tell whether it looks like a normal one or whether it is likely an anomaly or novelty.
- **Example:** Detecting unusual credit card transactions to prevent fraud.

Association Rule Learning

- In association rule learning, the goal is to dig into large amounts of data and discover interesting relations between attributes.
- **Example:** Suppose you own a supermarket. Running an association rule on your sales logs may reveal that people who purchase eggs, also tend to buy bread. Thus you may want to place these items close to each other.



Exercise

Of the following examples, which ones do you think are Unsupervised Learning problems as opposed to Supervised Learning problems? Also classify the sub-type.

1. Given data about the size of houses on the real estate market, try to predict their price.
- Supervised learning, Regression
2. Given data about house prices on a real estate market, predict if a house sells for more or less than the asking price.
- Supervised learning, Classification
3. Take a collection of 1,000,000 different genes, and find a way to automatically group these genes into groups that are somehow similar or related by different variables, such as lifespan, location, roles, etc.
- Unsupervised learning, Clustering
4. Given a picture of a person, we have to predict their age on the basis of the given picture.
- Supervised learning, Regression
5. You need to predict if people who purchase barbecue sauce and potato chips also tend to buy steak. Thus, you may want to place these items close to each other in your supermarket.
- Unsupervised learning, associative rule mining



Exercise

Of the following four examples, which ones, which of these four do you think would will be an Unsupervised Learning algorithm as opposed to Supervised Learning problem? Also classify the sub-type.

6. On Google News (<https://news.google.com>), when you search a particular news it shows you other related news along with it.
- Unsupervised learning, Clustering
7. We need an algorithm to catch manufacturing defects in products.
- Unsupervised learning, Anomaly detection
8. Given a patient with a tumor, we have to predict whether the tumor is malignant or benign.
- Supervised learning, Classification
9. Social network analysis, where given knowledge about which friends you email the most or given your Facebook friends or your Google+ circles, can automatically identify which are cohesive groups of friends, also which are groups of people that all know each other.
- Unsupervised learning, Clustering



Exercise

Of the following four examples, which ones, which of these four do you think would will be an Unsupervised Learning algorithm as opposed to Supervised Learning problem? Also classify the sub-type.

10. Market segmentation, where using huge databases of customer information, you automatically discover market segments and automatically group your customers into different market segments so that you can automatically and more efficiently sell or market your different market segments together.
- Unsupervised learning, Clustering
11. We need an algorithm that can automatically remove outliers from a dataset before feeding it to another learning algorithm.
- Unsupervised learning, Anomaly detection
12. Suppose you are working on weather prediction, and use a learning algorithm to predict tomorrow's temperature (in degrees Centigrade/Fahrenheit).
- Supervised learning, Regression
13. Organize large computer clusters in a data center to figure out which machines tend to work together and if you can put those machines together, you can make your data center work more efficiently.
- Unsupervised learning, Clustering



Exercise

Of the following four examples, which ones, which of these four do you think would will be an Unsupervised Learning algorithm as opposed to Supervised Learning problem? Also classify the sub-type.

14. Take a collection of 1000 essays written on the US Economy, and find a way to automatically group these essays into a small number of groups of essays that are somehow "similar" or "related".
- Unsupervised learning, Clustering
15. Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.
- Supervised learning, Classification

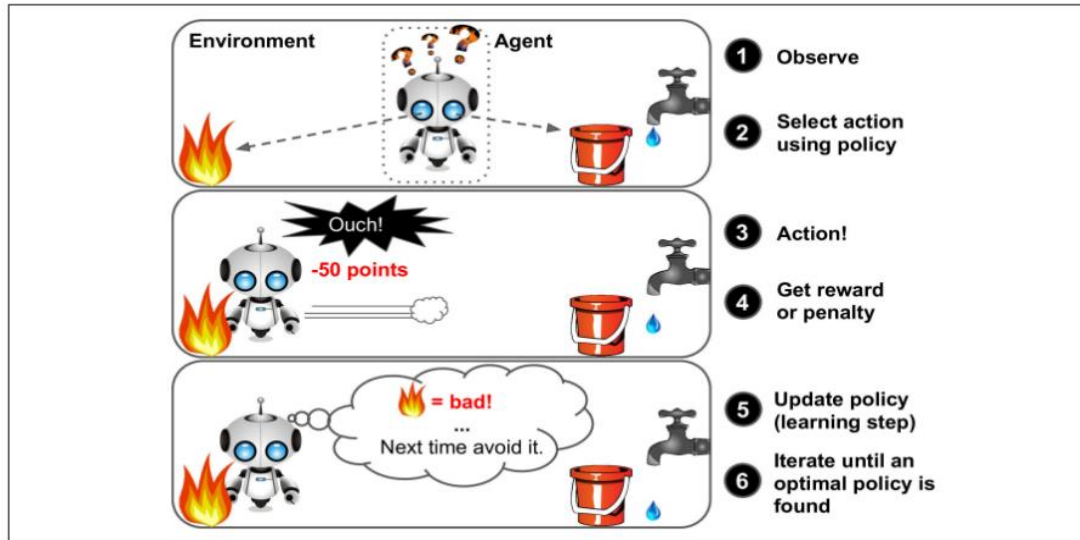


Semi-supervised Learning

- The agent learns from both supervised and unsupervised ways.
- These algorithms deal with partially labeled training data, usually a lot of unlabeled data and a little bit of labeled data.
- **Example:** In Google Photos, once you upload all your family photos to the service, it automatically recognizes that the same person A shows up in photos 1, 5, and 11, while another person B shows up in photos 2, 5, and 7. This is the unsupervised part of the algorithm (clustering). Now all the system needs is for you to tell it who these people are. Just one or a few labels per person and it is able to name everyone in every photo, which is useful for searching photos.

Reinforcement Learning

- The agent learns from a series of reinforcements: rewards and punishments.
- In general, a reinforcement learning agent is able to perceive and interpret its environment, take actions and learn through trial and error.
- It must then learn by itself what is the best strategy, called a policy, to get the most reward over time.
- A policy defines what action the agent should choose when it is in a given situation.
- **Example:** Many robots implement Reinforcement Learning algorithms to learn how to walk.





Instance-Based Versus Model-Based Learning

- One more way to categorize machine learning systems is by how they generalize.
- Most Machine Learning tasks are about making predictions.
- This means that given a number of training examples, the system needs to be able to generalize to examples it has never seen before.

Instance-based learning (Non-parametric Learning)

- Instance based system learns the examples by heart, then generalizes to new cases by comparing them to the learned examples (or a subset of them), using a similarity measure.
- **Example:** The system would flag an email as spam if it has many words in common with a known spam email, i.e. it will count the similar words to declare an email as spam.

Model-based learning (Parametric Learning)

- Model based system generalizes from a set of examples to build a model of these examples, then use that model to make predictions.
- **Example:** Fit a line equation on linear data and then use the equation of line of line for predictions.



Common Learning Algorithms

Commonly used supervised learning algorithms:

- Linear Regression (parametric, regression)
- Logistic Regression (parametric, classification)
- Naïve Bayes (non-parametric, regression, classification)
- Decision Trees (non-parametric, regression, classification)
- Random Forests (non-parametric, regression, classification)
- K Nearest Neighbors (KNN) (non-parametric, regression, classification)
- Support Vector Machines (SVM) (parametric, regression, classification)

Commonly used unsupervised learning algorithms:

- K-means Clustering (clustering)
- Hierarchical Cluster Analysis (clustering)
- One-class SVM (anomaly detection and novelty detection)
- Principal Component Analysis (PCA) (visualization and dimensionality reduction)
- Apriori Algorithm (association rule learning)



Main Challenges of Machine Learning

Insufficient Quantity of Training Data

- It takes a lot of data for most Machine Learning algorithms to work properly.
- Even for very simple problems you typically need thousands of examples, and for complex problems such as image or speech recognition you may need millions of examples.
- Some studies have shown that a simple algorithm can also perform as good as an advanced algorithm if sufficient data is there.
- For a toddler to learn what an apple is, all it takes is for you to point to an apple and say “apple” (possibly repeating this procedure a few times). Now the child is able to recognize apples in all sorts of colors and shapes. Genius. Machine Learning is not quite there yet.

Non-representative Training Data

- It is crucial to use a training set that is representative of the cases you want to generalize to.
- Non-representative data can cause sampling bias.



Main Challenges of Machine Learning

Poor-Quality Data

- Obviously, if your training data is full of errors, outliers, and noise, it will make it harder for the system to detect the underlying patterns, so your system is less likely to perform well.
- It is often well worth the effort to spend time cleaning up your training data.

Irrelevant Features

- Your system will only be capable of learning if the training data contains enough relevant features and not too many irrelevant ones.
- A critical part of the success of a machine learning project is coming up with a good set of features to train on. This process, called feature engineering.

Wrong Algorithm Selection

- Since a large variety of algorithms is available, choosing the right algorithm sometimes becomes a challenge.
- A wrong algorithm may cause overfitting or underfitting of data.



Thank you!